

Rosenberg Compact-Operator Model

[Earlier public Project view and PDF](#)

[Rosenberg reading routes](#) · [Reading routes PDF](#)

MATHLIBANNEX / THEOREM PROJECT

The mathematical goal

Let A be a nonzero complex C^* -algebra, not assumed unital, and let π act on a separable complex Hilbert space H as a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . Then π is injective, every represented operator is compact, and every compact operator on H has an exact preimage in A .

Exact source: [MathlibAnnex v0.4.0 Project entry](#) · [Source manifest](#)

Thirty-three Declaration Card expositions are available. The added density route is source-only and creates no Card.

Why it matters

The result identifies A , through π , with the compact-operator model in an exact unbundled sense. The Project exposes the reusable proof route through faithfulness, character eigenvectors, countability, scalar corners, rank-one preimages and compact approximation.

Hypotheses

A is nonzero and may be genuinely nonunital. H is separable. The representation π is nonzero and irreducible, and every nonzero irreducible representation of A is unitarily equivalent to π . No separability of A , prior faithfulness or simplicity is assumed. The unital finite-dimensional consequences have their own exact hypotheses.

Scope limits

The 33 Card explanations have completed source-exposition correspondence review against exact MathlibAnnex v0.4.0 source. The compact-operator characterization is unbundled; no bundled StarAlgEquiv is supplied. The original 1953 paper and the Farah book remain unacquired and unreviewed in this record.

Proof architecture

Faithfulness

A pure state detecting a nonzero kernel element yields a nonzero irreducible GNS representation. Because every nonzero irreducible representation lies in the unique unitary-equivalence class represented by π , unitary equivalence forces the kernel element to act nontrivially, a contradiction.

[Faithfulness of a representative of the unique irreducible-representation class](#)

Characters and eigenvectors

A character on a closed star subalgebra of the minimal unitization extends to a pure state. For a character distinct from the scalar character, the restricted GNS representation produces a unit joint eigenvector with that prescribed character.

[Eigenvector for a character distinct from the scalar character](#)

Countability and a scalar corner

Distinct character eigenvectors are orthogonal. Separability of H makes the nonscalar character space countable; an isolated character then yields a nonzero projection whose corner is scalar.

[Countability of the full character space · Scalar corner from a representative of the unique irreducible-representation class](#)

Every compact operator has a preimage

A rank-one image and irreducible interpolation give exact preimages for operators of the form rankOne . Closed range and finite-rank approximation then give an exact preimage for every compact operator.

[Preimages of every compact operator](#)

Every represented operator is compact

If some represented operator were noncompact, a character would separate a maximal abelian witness from the compact-preimage ideal. The resulting eigenvector conflicts with a rank-one preimage, so every image operator is compact.

[Compact image of a representative of the unique irreducible-representation class](#)

Unital finite-dimensional consequences

For a unital algebra, the version quantifying competitors through the nonunital representation interface yields the same unique-class condition. Compactness of the identity image forces H to be finite-dimensional, then injectivity forces A to be finite-dimensional, giving the stated nonexistence consequence.

[Finite-dimensional algebra from a separable representative of the unique irreducible-representation class](#)

Density below the continuum and the compact-operator model

The compact-operator model route extends from separable representation spaces to an irreducible representation space with a norm-dense subset of cardinality strictly below the continuum. The nonunital statements retain the ordinary singleton-model hypotheses. Contrapositively, a singleton model that is not a compact-operator model forces every norm-dense subset of its representation

space, and every norm-dense subset of the algebra, to have cardinality at least the continuum. This is a source-only extension of the reviewed 33-Card route; it creates no additional Cards. These statements are about the singleton irreducible model; the lower bound on its Hilbert space does not apply to arbitrary faithful reducible representations.

45 selected declarations

[isCompactOperatorModel_of_singleton_of_dense · isCompactOperatorModel_of_singleton_of_dense_algebra · continuum_le_cardinalMk_dense_space_of_singleton_of_not_isCompactOperatorModel · continuum_le_cardinalMk_dense_algebra_of_singleton_of_not_isCompactOperatorModel](#)

Boundary Inputs

2690 exact Boundary Inputs: EXTERNAL_COMPILED_DECLARATION 2233, OMITTED_INTERNAL_NATIVE_DECLARATION 457. Selected reachability is retained through every omitted internal declaration.

▼ Inspect Boundary Inputs

- Eq — Compiled external provider used by selected Project declarations.
- Membership.mem — Compiled external provider used by selected Project declarations.
- PseudoMetricSpace.toUniformSpace — Compiled external provider used by selected Project declarations.
- OfNat.ofNat — Compiled external provider used by selected Project declarations.
- UniformSpace.toTopologicalSpace — Compiled external provider used by selected Project declarations.
- And — Compiled external provider used by selected Project declarations.
- Complex — Compiled external provider used by selected Project declarations.
- Exists — Compiled external provider used by selected Project declarations.
- Semiring.toNonAssocSemiring — Compiled external provider used by selected Project declarations.
- Complex.instNormedField — Compiled external provider used by selected Project declarations.
- Ring.toSemiring — Compiled external provider used by selected Project declarations.
- Semiring.toMonoid — Compiled external provider used by selected Project declarations.
- Set — Compiled external provider used by selected Project declarations.
- AddCommMonoid.toAddMonoid — Compiled external provider used by selected Project declarations.
- AddGroup.toSubNegMonoid — Compiled external provider used by selected Project declarations.

Dependency-first reading route

Levels belong to this Project. Select a level or follow a relation to another declaration tile.

Level 0 16 declarations

nonempty_iInter_of_isCompact`CantorScheme.nonempty_iInter_of_isCompact`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT[exists_injective_nat_bool_of_isTopologicalBasis_clopens](#)[Read exact source](#)**A compact-operator model for a representation**`MathlibAnnex.Analysis.CStarAlgebra.IsCompactOperatorModel`

Packages injectivity, compact-valuedness, and surjectivity onto all compact operators into one unbundled predicate.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital, H a complex Hilbert space, and e a nonunital complex star-algebra homomorphism from A to the bounded operators on H . The model predicate consists of all three clauses: e is injective; every represented $e(a)$ is compact; and each compact bounded operator T on H equals $e(a)$ for some a in A .

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[Compact-operator model from a representative of the unique irreducible-representation class,](#)
[isCompactOperatorModel_of_singleton_of_dense,](#)
[Compact-operator model from a unital representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL countability-scalar-corner · density · every-compact ·
0 only-compact · rosenberg · unital-consequences

Maximal abelian *-subalgebras

MathlibAnnex.Analysis.CStarAlgebra.IsMaximalAbelian

Defines maximal abelian unital star subalgebras by commutativity and maximality under inclusion.

▼ Statement in Project context

Let D be a unital star subalgebra of a unital complex C^* -algebra A . D is commutative, and any commutative unital star subalgebra E containing D is contained in D ; thus D is maximal for inclusion among such subalgebras.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[exists_nonzero_projection_scalar_corner_of_isolated_character_ne_infinity](#),
[mem_maximalAbelian_of_map_eq_rankOne_of_eigenvector](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)

▼ Show 1 more

[Maximal abelian subalgebra containing a self-adjoint element](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL characters-eigenvectors · countability-scalar-corner ·
0 density · every-compact · faithfulness · only-compact ·
rosenberg · unital-consequences

Pure states as real extreme points

MathlibAnnex.Analysis.CStarAlgebra.IsPureState

Defines pure states as the real extreme points of the state space.

▼ Statement in Project context

Let A be a unital complex C^* -algebra and let $\varphi : A \rightarrow \mathbb{C}$ be a continuous complex-linear functional. φ is pure exactly when it belongs to the real extreme points of the normalized positive state space $\text{stateSpace } A$. Membership in that extreme-point set includes membership in the state space.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[Pure-state extension of a character](#), [Pure-state GNS irreducibility](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0 characters-eigenvectors · countability-scalar-corner · density · every-compact · faithfulness · only-compact · rosenberg

Representations without a unit-preservation requirement

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation

Names nonunital complex star-algebra homomorphisms into bounded operators as the representation type.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital, and H a complex Hilbert space. A representation of A on H is a complex-linear nonunital star-algebra homomorphism into the bounded complex-linear operators on H . It preserves addition, multiplication, complex scalar multiplication and involution; neither preservation of a unit nor faithfulness is built in.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[Nonunital irreducibility](#), [Unitary equivalence of nonunital representations](#),
[character_apply_eq_one_of_map_eq_rankOne](#)

▼ Show 2 more

[exists_character_ne_infinity_annihilating_compact_preimage](#),
[mem_maximalAbelian_of_map_eq_rankOne_of_eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0 density

exists_character_ne_infinity_apply_ne_zero

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_character_ne_infinity_apply_ne_zero

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[exists_isolated_character_ne_infinity_of_cardinalMk_It_cont_inuum](#)

[Read exact source](#)

LEVEL 0

rosenberg

The scalar character of the unitization

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.infinityCharacter`

Defines the distinguished scalar-coordinate character of the minimal unitization.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital, and form its minimal unitization $\text{Unitization } \mathbb{C} A$. The infinity character is the character induced by projection onto the scalar coordinate: it evaluates a unitization element at its $\mathbb{C}$ component. It is a character of $\text{Unitization } \mathbb{C} A$, not of A itself.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

None in this Project

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

rosenberg · unital-consequences

Singleton condition tested against all irreducible representations

`MathlibAnnex.Analysis.CStarAlgebra.Representation.IsSingletonIrreducibleModelAmongNonUnital`

Defines the corresponding condition for a unital representation, with competing nonzero irreducible representations quantified through the nonunital representation interface.

▼ Statement in Project context

Let A be a unital complex C^* -algebra, H a complex Hilbert space, and π a unital representation of A on H . The representation π is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A : π is irreducible, and for every nonzero irreducible representation ρ presented through the nonunital representation interface, π is unitarily equivalent to the associated unital representation $\rho.\text{toUnital}$. Faithfulness and separability are not part of the definition.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[Compact-operator model from a unital representative of the unique irreducible-representation class](#), [Finite-dimensional algebra from a separable representative of the unique irreducible-representation class](#), [Finite-dimensional space for a separable representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

density

inner_eq_zero_of_ne_characters

`MathlibAnnex.Analysis.CStarAlgebra.Representation.inner_eq_zero_of_ne_characters`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[cardinalMk_nonScalarCharacterSpace_lt_continuum_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 0

density

one_le_dist_of_norm_eq_one_of_inner_eq_zero

`MathlibAnnex.Analysis.CStarAlgebra.Representation.one_le_dist_of_norm_eq_one_of_inner_eq_zero`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[cardinalMk_nonScalarCharacterSpace_lt_continuum_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 0

rosenberg

Full operator image in finite dimension`MathlibAnnex.Analysis.CStarAlgebra.Representation.surjective_of_irreducible_of_finiteDimensional`

Shows that an irreducible representation on a nonzero finite-dimensional Hilbert space has full operator image.

▼ Statement in Project context

Let A be a unital complex C^* -algebra, H a nonzero finite-dimensional complex Hilbert space, and π an irreducible unital representation of A on H . No singleton hypothesis or separability assumption on A is made. π is surjective onto all bounded complex-linear operators on H .

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[Compact-operator model from a unital representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

density · only-compact · rosenberg

A character separating a closed ideal`MathlibAnnex.Analysis.CStarAlgebra.exists_character_annihilating_closedIdeal_of_not_mem`

Separates an element from a closed ideal of a commutative C^* -algebra by a character vanishing on the ideal.

▼ Statement in Project context

Let D be a commutative unital complex C^* -algebra, J an ideal whose carrier is norm-closed, and $d \in D$ with $d \notin J$. There is a character χ of D that vanishes on every $x \in J$ while $\chi(d) \neq 0$. Properness follows from $d \notin J$ and is not separately assumed.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[exists_character_ne_infinity_annihilating_compact_preimage](#), [Compact image of a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

density · only-compact · rosenberg · unital-consequences

The preimage ideal of the compact operators`MathlibAnnex.CStarAlgebra.compactPreimageIdeal`

Collects the elements represented by compact operators into a two-sided algebraic ideal.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital, H a complex Hilbert space, and ρ a representation of A on H . `compactPreimageIdeal` ρ is the two-sided algebraic ideal of elements whose represented bounded operator is compact. Norm-closedness is a separate theorem and is not a field of this returned `TwoSidedIdeal`.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[exists_character_ne_infinity_annihilating_compact_preimage](#), [Compact image of a representative of the unique irreducible-representation class](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 0

density

denseRange_restrict_of_continuous`MathlibAnnex.Topology.denseRange_restrict_of_continuous`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[exists_dense_cardinalMk_It_continuum_of_continuous_denseRange](#)

[Read exact source](#)

LEVEL 0

density

exists_injective_into_dense_of_separated`MathlibAnnex.Topology.exists_injective_into_dense_of_separated`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[cardinalMk_le_cardinalMk_dense_of_separated](#)

[Read exact source](#)

LEVEL 0

countability-scalar-corner · every-compact · only-compact · rosenberg · unital-consequences

An isolated point in a countable Baire space`MathlibAnnex.Topology.exists_isOpen_singleton`

Extracts an isolated point from a nonempty countable T1 Baire space.

▼ Statement in Project context

Let X be nonempty, countable, T1 and Baire. At least one singleton $\{x\}$ is open in X .

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this Project

USED BY IN THIS PROJECT

[Isolated character away from infinity](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 13 declarations**LEVEL 1**

characters-eigenvectors · countability-scalar-corner · density · every-compact · faithfulness · only-compact · rosenberg

Nonzero irreducibility without a unit assumption`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible`

Defines nonunital irreducibility as nonzero action with no nontrivial closed reducing subspaces.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital, H a complex Hilbert space, and π a representation of A on H . The predicate says that π acts nontrivially and every closed complex submodule L invariant under both $\pi(a)$ and its adjoint for every a is either $\{0\}$ or H .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Possibly nonunital representation](#)

USED BY IN THIS PROJECT

[Representative of the unique irreducible-representation class](#),
[denseRange_apply_of_isIrreducible_of_apply_ne_zero](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

characters-eigenvectors · countability-scalar-corner · density · every-compact · faithfulness · only-compact · rosenberg

Unitary equivalence of two representations`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

Defines unitary equivalence by an onto complex-linear isometry intertwining the two representations.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital; let π and ρ be representations of A on complex Hilbert spaces H and K , respectively. They are unitarily equivalent when an onto complex-linear isometry U from H to K intertwines every represented operator at every vector.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Possibly nonunital representation](#)

USED BY IN THIS PROJECT

[Representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

density

character_apply_eq_one_of_map_eq_rankOne

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.character_apply_eq_one_of_map_eq_rankOne

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Possibly nonunital representation](#)

USED BY IN THIS PROJECT

[isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

[Read exact source](#)

LEVEL 1

density

exists_character_ne_infinity_annihilating_compact_preimage

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_character_ne_infinity_annihilating_compact_preimage

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Possibly nonunital representation](#), [A character separates a closed ideal](#), [Algebraic compact-preimage ideal](#)

USED BY IN THIS PROJECT

[isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

[Read exact source](#)

LEVEL 1

countability-scalar-corner · every-compact · only-compact · rosenberg

An isolated character away from the scalar character

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_isolated_character_ne_infinity

Finds an isolated character distinct from the scalar character in a countable character space.

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital. Let D be a closed commutative unital star subalgebra of Unitization $\mathbb{C}A$ with countable character space, and let $0 \neq d \in D$ have zero scalar coordinate. There exists a character χ distinct from the restricted infinity character for which $\{\chi\}$ is open in the character space. This theorem itself does not assume that any representation represents a unique irreducible-representation class.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[An isolated point in a countable Baire space](#)

USED BY IN THIS PROJECT

[Scalar corner from a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

density

exists_nonzero_projection_scalar_corner_of_isolated_character_ne_infinity

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner_of_isolated_character_ne_infinity

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Maximal abelian star subalgebra](#)

USED BY IN THIS PROJECT

[exists_nonzero_projection_scalar_corner_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 1

density

**mem_maximalAbelian_of_map_eq_rank
One_of_eigenvector**

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.mem_maximalAbelian_of_map_eq_rankOne_of_eigenvector

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Maximal abelian star subalgebra](#), [Possibly nonunital representation](#)

USED BY IN THIS PROJECT

[isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

[Read exact source](#)

LEVEL

countability-scalar-corner · density · every-compact · only-compact · rosenberg

**A maximal abelian subalgebra containing
a self-adjoint element**

MathlibAnnex.Analysis.CStarAlgebra.exists_maximalAbelian_containing_isSelfAdjoint

Places a self-adjoint element inside a maximal abelian unital star subalgebra.

▼ Statement in Project context

Let x be self-adjoint in a unital complex C^* -algebra A . There exists a maximal abelian unital star subalgebra D of A containing x .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Maximal abelian star subalgebra](#)

USED BY IN THIS PROJECT

[Scalar corner from a representative of the unique irreducible-representation class](#), [exists_nonzero_projection_scalar_corner_of_singleton_of_dense](#), [isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL

1 characters-eigenvectors · countability-scalar-corner · density · every-compact · faithfulness · only-compact · rosenberg · unital-consequences

A character extends to a pure state

MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension

Extends a character of a closed unital star subalgebra to a pure state of the ambient algebra.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra, D a closed unital star subalgebra of A , and χ a character of D . There is a continuous complex-linear functional φ on A in `stateSpace A` that is pure and restricts to χ : for every d in D , $\varphi(d) = \chi(d)$.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Pure state as an extreme point](#)

USED BY IN THIS PROJECT

[Eigenvector for a character distinct from the scalar character](#), [Pure state detecting a nonzero square](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL

1 characters-eigenvectors · countability-scalar-corner · density · every-compact · faithfulness · only-compact · rosenberg · unital-consequences

**The GNS representation of a pure state
is irreducible**

MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom

Turns purity of a state into irreducibility of its canonical GNS representation.

▼ Statement in Project context

Let A be a unital complex C^* -algebra and let φ be a pure state of A . The canonical GNS star-algebra homomorphism built from the positive map associated to φ is irreducible.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Pure state as an extreme point](#)

USED BY IN THIS PROJECT

[Eigenvector for a character distinct from the scalar character](#), [Faithfulness of a representative of the unique irreducible-representation class](#), [Simplicity from a faithful representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 1

density

cardinalMk_le_cardinalMk_dense_of_separated

MathlibAnnex.Topology.cardinalMk_le_cardinalMk_dense_of_separated

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_injective_into_dense_of_separated](#)

USED BY IN THIS PROJECT

[cardinalMk_lt_continuum_of_separated_of_dense](#)

[Read exact source](#)

LEVEL 1

density

exists_dense_cardinalMk_lt_continuum_of_continuous_denseRange

MathlibAnnex.Topology.exists_dense_cardinalMk_lt_continuum_of_continuous_denseRange

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[denseRange_restrict_of_continuous](#)

USED BY IN THIS PROJECT

[exists_dense_cardinalMk_lt_continuum_of_isIrreducible](#)

[Read exact source](#)

LEVEL 1

density

exists_injective_nat_bool_of_isTopologicalBasis_clopens

MathlibAnnex.Topology.exists_injective_nat_bool_of_isTopologicalBasis_clopens

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[nonempty_iInter_of_isCompact](#)

USED BY IN THIS PROJECT

[exists_isOpen_singleton_of_compactSpace_of_cardinalMk_lt_continuum](#)

[Read exact source](#)

Level 2 6 declarations

A representative of the unique irreducible class

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel`

Defines when a nonzero irreducible representation π is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A .

▼ Statement in Project context

Let A be a complex C^* -algebra, not assumed unital, H a complex Hilbert space, and π a representation of A on H . The representation π is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A : π is nonzero and irreducible, and every nonzero irreducible representation of A on any complex Hilbert space is unitarily equivalent to π . The definition does not assume faithfulness or separability of A or H .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Nonunital irreducibility](#), [Unitary equivalence of nonunital representations](#)

USED BY IN THIS PROJECT

[Eigenvector for a character distinct from the scalar character](#), [Faithfulness of a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

denseRange_apply_of_isIrreducible_of_apply_ne_zero

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.denseRange_apply_of_isIrreducible_of_apply_ne_zero`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Nonunital irreducibility](#)

USED BY IN THIS PROJECT

[exists_denseRange_apply_of_isIrreducible](#)

[Read exact source](#)

LEVEL 2

rosenberg · unital-consequences

A faithful singleton model forces simplicity`MathlibAnnex.Analysis.CStarAlgebra.Representation.isSimpleCStarAlgebra_of_singleton_of_injective`

Converts a faithful representative of the unique irreducible-representation class into closed-two-sided-ideal simplicity of the algebra.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra, H a complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . Assume additionally that π is injective. No separability assumption on H is required. A is simple in the exact closed-two-sided-ideal sense `IsSimpleCStarAlgebra A`.

IMMEDIATE PREREQUISITES IN THIS PROJECT[Pure-state GNS irreducibility](#)**USED BY IN THIS PROJECT**[Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 2**

density · every-compact · faithfulness · only-compact · rosenberg · unital-consequences

A pure state detects a nonzero square`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_nonzero_on_star_mul_self`

Supplies a pure state that detects a^*a , enabling faithfulness arguments without separability of the algebra.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra and let $a \in A$ be nonzero. No separability assumption on A is required. Some pure state ϕ in `stateSpace A` satisfies $\phi(a^*a) \neq 0$.

IMMEDIATE PREREQUISITES IN THIS PROJECT[Pure-state extension of a character](#)**USED BY IN THIS PROJECT**[Faithfulness of a representative of the unique irreducible-representation class](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)[Open Declaration Card](#) [Read exact source](#)**LEVEL 2**

density

cardinalMk_lt_continuum_of_separated_of_dense`MathlibAnnex.Topology.cardinalMk_lt_continuum_of_separated_of_dense`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT[cardinalMk_le_cardinalMk_dense_of_separated](#)**USED BY IN THIS PROJECT**[cardinalMk_nonScalarCharacterSpace_lt_continuum_of_singleton_of_dense](#)[Read exact source](#)**LEVEL 2**

density

exists_isOpen_singleton_of_compactSpace_of_cardinalMk_lt_continuum`MathlibAnnex.Topology.exists_isOpen_singleton_of_compactSpace_of_cardinalMk_lt_continuum`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT[exists_injective_nat_bool_of_isTopologicalBasis_clopens](#)**USED BY IN THIS PROJECT**[exists_isOpen_singleton_of_cardinalMk_lt_continuum](#)[Read exact source](#)**Level 3** 5 declarations

LEVEL 3

density

exists_denseRange_apply_of_isIrreducible

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_denseRange_apply_of_isIrreducible

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[denseRange_apply_of_isIrreducible_of_apply_ne_zero](#)

USED BY IN THIS PROJECT

[exists_dense_cardinalMk_It_continuum_of_isIrreducible](#)

[Read exact source](#)

LEVEL**3**

characters-eigenvectors · countability-scalar-corner · density · every-compact · only-compact · rosenberg

A character becomes a joint unit eigenvector

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_unit_eigenvector_of_character_ne_infinity

Produces a unit joint eigenvector whose eigenvalue character is the prescribed character distinct from the scalar character.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital; let H be a complex Hilbert space; let π be a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A ; let D be a closed unital star subalgebra of $\text{Unitization } \mathbb{C} A$; and let χ be a character of D distinct from $\text{infinityCharacterOn } D$. No separability assumption on H is required. There is a unit vector η in H such that $\pi.\text{unitization}(d)\eta = \chi(d)\eta$ for every d in D .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Representative of the unique irreducible-representation class](#), [Pure-state extension of a character](#), [Pure-state GNS irreducibility](#)

USED BY IN THIS PROJECT

[cardinalMk_nonScalarCharacterSpace_It_continuum_of_singleton_of_dense](#), [Countability of the full character space](#), [isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

[Open Declaration Card](#) [Read exact source](#)

Faithfulness from a unique irreducible class

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton`

Shows that a representative of the unique irreducible-representation class of a nonzero C^* -algebra is faithful.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital, H a complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . No separability assumption on H is required. Then π is injective.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Representative of the unique irreducible-representation class](#), [Pure state detecting a nonzero square](#), [Pure-state GNS irreducibility](#)

USED BY IN THIS PROJECT

[A represented rank-one projection from a representative of the unique irreducible-representation class](#), [exists_norm_eq_one_and_map_eq_rankOne_of_singleton_of_dense](#), [isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

[Open Declaration Card](#) [Read exact source](#)

A unital singleton model acts in finite dimension

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton`

Shows that the Hilbert space of a separably acting unital representative of the unique irreducible-representation class is finite-dimensional.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra, H a separable complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . H is finite-dimensional over $\mathbb{C}$. The conclusion is conditional on the unique-class hypothesis, not a statement about all unital irreducible representations.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Maximal abelian star subalgebra](#), [Simplicity from a faithful representative of the unique irreducible-representation class](#), [Pure state detecting a nonzero square](#)

▼ Show 2 more

[Algebraic compact-preimage ideal](#), [An isolated point in a countable Baire space](#)

USED BY IN THIS PROJECT

[Compact-operator model from a unital representative of the unique irreducible-representation class](#), [Finite-dimensional algebra from a separable representative of the unique irreducible-representation class](#), [Finite-dimensional space for a separable representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 3

density

exists_isOpen_singleton_of_cardinalMk_lt_continuum

MathlibAnnex.Topology.exists_isOpen_singleton_of_cardinalMk_lt_continuum

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_isOpen_singleton_of_compactSpace_of_cardinalMk_lt_continuum](#)

USED BY IN THIS PROJECT

[exists_isolated_character_ne_infinity_of_cardinalMk_lt_continuum](#)

[Read exact source](#)

Level 4 8 declarations**LEVEL 4**

density

cardinalMk_nonScalarCharacterSpace_lt_continuum_of_singleton_of_dense

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.cardinalMk_nonScalarCharacterSpace_lt_continuum_of_singleton_of_dense

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Eigenvector for a character distinct from the scalar character, inner_eq_zero_of_ne_characters, one_le_dist_of_norm_eq_one_of_inner_eq_zero](#)

▼ Show 1 more

[cardinalMk_lt_continuum_of_separated_of_dense](#)

USED BY IN THIS PROJECT

[exists_nonzero_projection_scalar_corner_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 4

countability-scalar-corner · every-compact · only-compact · rosenberg

The full character space is countable

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.countable_characterSpace_of_nonUnital_singleton

Shows that the full character space of a closed unitization subalgebra is countable when π is a separably acting representative of the unique irreducible-representation class.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital; let H be a separable complex Hilbert space; let π be a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A ; and let D be a closed unital star subalgebra of Unitization $\odot A$. The entire character space of D is countable, including the distinguished scalar character.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Eigenvector for a character distinct from the scalar character](#)

USED BY IN THIS PROJECT

[Scalar corner from a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 4

density

exists_dense_cardinalMk_lt_continuum_of_isIrreducible

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_dense_cardinalMk_lt_continuum_of_isIrreducible

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_denseRange_apply_of_isIrreducible](#),
[exists_dense_cardinalMk_lt_continuum_of_continuous_denseRange](#)

USED BY IN THIS PROJECT

[isCompactOperatorModel_of_singleton_of_dense_algebra](#)

[Read exact source](#)

LEVEL 4

density

exists_isolated_character_ne_infinity_of_cardinalMk_lt_continuum

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_isolated_character_ne_infinity_of_cardinalMk_lt_continuum

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_character_ne_infinity_apply_ne_zero](#),
[exists_isOpen_singleton_of_cardinalMk_lt_continuum](#)

USED BY IN THIS PROJECT

[exists_nonzero_projection_scalar_corner_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 4

density

isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[character_apply_eq_one_of_map_eq_rankOne](#),
[exists_character_ne_infinity_annihilating_compact_preimage](#), [Eigenvector for a character distinct from the scalar character](#)

▼ Show 3 more

[Faithfulness of a representative of the unique irreducible-representation class](#),
[mem_maximalAbelian_of_map_eq_rankOne_of_eigenvector](#), [Maximal abelian subalgebra containing a self-adjoint element](#)

USED BY IN THIS PROJECT

[isCompactOperator_map_of_singleton_of_rankOne_preimages](#)

[Read exact source](#)

LEVEL 4

rosenberg

Compact-operator conclusion for the unital singleton condition

MathlibAnnex.Analysis.CStarAlgebra.Representation.f_aithful_and_compactOperatorModel_of_singleton_amongNonUnital

Shows that a separably acting unital representative of the unique irreducible-representation class is faithful and has image exactly $K(H)$.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra, H a separable complex Hilbert space, and π a unital representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . π is injective; its underlying nonunital star homomorphism is injective, maps every $a \in A$ to a compact operator, and represents every compact operator on H .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Unbundled compact-operator model](#), [Unital representative of the unique irreducible-representation class](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)

▼ Show 1 more

[Full operator image in finite dimension](#)

USED BY IN THIS PROJECT

None in this Project

[Open Declaration Card](#) [Read exact source](#)

Finite-dimensional algebra in the unital singleton case

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_algebra_of_singleton_amongNonUnital`

Shows that a nonzero unital C^* -algebra is finite-dimensional when it has a representative of its unique irreducible-representation class on a separable Hilbert space.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra, H a separable complex Hilbert space, and π a unital representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . Then A is finite-dimensional over $\mathbb{C}$; finite dimension of A is the conclusion, not an initial hypothesis.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Unital representative of the unique irreducible-representation class](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

[No separably acting representative of all nonzero irreducible representations](#)

[Open Declaration Card](#) [Read exact source](#)

Finite-dimensional representation space in the unital singleton case

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton_amongNonUnital`

Shows that the Hilbert space of a separably acting representative of the unique irreducible-representation class is finite-dimensional.

▼ Statement in Project context

Let A be a nonzero unital complex C^* -algebra, H a separable complex Hilbert space, and π a unital representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . Then H is finite-dimensional over $\mathbb{C}$.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Unital representative of the unique irreducible-representation class](#), [Finite-dimensional space for a unital representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

None in this Project

[Open Declaration Card](#) [Read exact source](#)

Level 5 4 declarations

LEVEL
5

countability-scalar-corner · every-compact · only-compact · rosenberg

A nonzero projection with scalar corner

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner`

Produces a nonzero projection with one-dimensional corner from a separably acting representative of the unique irreducible-representation class.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital, H a separable complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . There is a nonzero star projection p in A whose every corner $p a p$ is a complex scalar multiple of p .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Countability of the full character space](#), [Isolated character away from infinity](#), [Maximal abelian subalgebra containing a self-adjoint element](#)

USED BY IN THIS PROJECT

[A represented rank-one projection from a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

LEVEL 5

density

exists_nonzero_projection_scalar_corner_of_singleton_of_dense

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner_of_singleton_of_dense`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[cardinalMk_nonScalarCharacterSpace_of_continuum_of_singleton_of_dense](#),
[exists_isolated_character_ne_infinity_of_cardinalMk_of_continuum](#),
[exists_nonzero_projection_scalar_corner_of_isolated_character_ne_infinity](#)

▼ Show 1 more

[Maximal abelian subalgebra containing a self-adjoint element](#)

USED BY IN THIS PROJECT

[exists_norm_eq_one_and_map_eq_rankOne_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 5

density

isCompactOperator_map_of_singleton_of_rankOne_preimages

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton_of_rankOne_preimages

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[isCompactOperator_map_of_isSelfAdjoint_of_singleton_of_rankOne_preimages](#)

USED BY IN THIS PROJECT

[isCompactOperatorModel_of_singleton_of_dense](#)

[Read exact source](#)

LEVEL 5

rosenberg

An infinite-dimensional unital algebra has no separable singleton model

MathlibAnnex.Analysis.CStarAlgebra.Representation.not_singleton_amongNonUnital_of_infiniteDimensional

Shows that, for an infinite-dimensional unital C^* -algebra, no representation on a separable Hilbert space is a representative of the unique unitary-equivalence class of all nonzero irreducible representations.

▼ Statement in Project context

Let A be a nonzero infinite-dimensional unital complex C^* -algebra, H a separable complex Hilbert space, and π a unital representation of A on H . The representation π is not a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . Equivalently, π is not both irreducible and unitarily equivalent to every nonzero irreducible representation of A . This does not rule out separable irreducible representations in general.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Finite-dimensional algebra from a separable representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

None in this Project

[Open Declaration Card](#) [Read exact source](#)

Level 6 2 declarations

A projection represented by a rank-one operator

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_rankOne_map`

Produces a nonzero projection whose represented image is a compact rank-one orthogonal projection.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital, H a separable complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . There are a nonzero star projection $p \in A$ and a unit vector $e \in H$ with $\pi(p) = \text{rankOne } e$; the represented operator $\pi(p)$ is compact. Minimality is not an explicit clause of this theorem type.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Scalar corner from a representative of the unique irreducible-representation class](#), [Faithfulness of a representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

[Preimages for all operators of the form `rankOne`](#)

[Open Declaration Card](#) [Read exact source](#)

`exists_norm_eq_one_and_map_eq_rankOne_of_singleton_of_dense`

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_norm_eq_one_and_map_eq_rankOne_of_singleton_of_dense`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_nonzero_projection_scalar_corner_of_singleton_of_dense](#), [Faithfulness of a representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

[exists_preimage_rankOne_of_singleton_of_dense](#)

[Read exact source](#)

Level 7 2 declarations

Every rank-one operator has an algebra preimage

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton`

Provides preimages for every operator of the form $\text{rankOne } \mathbb{C} \times y$, the rank-at-most-one generators used to obtain all compact operators.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital; let H be a separable complex Hilbert space; and let π be a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . For every pair $x, y \in H$, including zero vectors, there exists $a \in A$ with $\pi(a) = \text{rankOne } \mathbb{C} \times y$; this operator has rank at most one.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[A represented rank-one projection from a representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

[Preimages of every compact operator](#), [Compact image of a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

exists_preimage_rankOne_of_singleton_of_dense

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton_of_dense`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_norm_eq_one_and_map_eq_rankOne_of_singleton_of_dense](#)

USED BY IN THIS PROJECT

[isCompactOperatorModel_of_singleton_of_dense](#)

[Read exact source](#)

Level 8 3 declarations

Compact operators lie in a singleton representation range

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_of_compact_singleton`

Establishes the inclusion $K(H) \subseteq \pi(A)$ by giving a preimage for every compact operator.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital; let H be a separable complex Hilbert space; let π be a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A ; and let T be a compact bounded operator on H . There exists $a \in A$ with $\pi(a) = T$. This is the inclusion $K(H) \subseteq \pi(A)$; compactness of each $\pi(a)$ is proved separately.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Preimages for all operators of the form rankOne](#)

USED BY IN THIS PROJECT

[Compact-operator model from a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

isCompactOperatorModel_of_singleton_of_dense

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperatorModel_of_singleton_of_dense`

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Unbundled compact-operator model](#),
[exists_preimage_rankOne_of_singleton_of_dense](#),
[isCompactOperator_map_of_singleton_of_rankOne_preimages](#)

USED BY IN THIS PROJECT

[continuum_le_cardinalMk_dense_space_of_singleton_of_not_isCompactOperatorModel](#),
[isCompactOperatorModel_of_singleton_of_dense_algebra](#)

[Read exact source](#)

LEVEL 8

only-compact · rosenberg

Every operator in a singleton image is compact

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton`

Shows that every operator in the image of a separably acting representative of the unique irreducible-representation class is compact.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital, H a separable complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . No simplicity assumption is made. Every operator $\pi(a)$, $a \in A$, is compact.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Preimages for all operators of the form rankOne](#), [A character separates a closed ideal](#), [Algebraic compact-preimage ideal](#)

USED BY IN THIS PROJECT

[Compact-operator model from a representative of the unique irreducible-representation class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 9 3 declarations

LEVEL 9

density

continuum_le_cardinalMk_dense_space_of_singleton_of_not_isCompactOperatorModel

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.continuum_le_cardinalMk_dense_space_of_singleton_of_not_isCompactOperatorModel

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[isCompactOperatorModel_of_singleton_of_dense](#)

USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

LEVEL 9

rosenberg

A singleton model is faithful and exactly compact-valued

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.faithful_and_compactOperatorModel_of_singleton

Shows that a representative of the unique irreducible-representation class on a separable Hilbert space is faithful and has image exactly $K(H)$.

▼ Statement in Project context

Let A be a nonzero complex C^* -algebra, not assumed unital, H a separable complex Hilbert space, and π a representation of A on H that is a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . No separability of A , prior faithfulness, or simplicity is assumed. π is injective and satisfies the full unbundled compact-operator model: every $\pi(a)$ is compact and every compact T on H has an exact preimage in A .

IMMEDIATE PREREQUISITES IN THIS PROJECT

[Unbundled compact-operator model](#), [Preimages of every compact operator](#), [Compact image of a representative of the unique irreducible-representation class](#)

USED BY IN THIS PROJECT

None in this Project

[Open Declaration Card](#) [Read exact source](#)

LEVEL 9

density

isCompactOperatorModel_of_singleton_of_dense_algebra

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperatorModel_of_singleton_of_dense_algebra

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[exists_dense_cardinalMk_lt_continuum_of_isIrreducible](#), [isCompactOperatorModel_of_singleton_of_dense](#)

USED BY IN THIS PROJECT

[continuum_le_cardinalMk_dense_algebra_of_singleton_of_not_isCompactOperatorModel](#)

[Read exact source](#)

Level 10 1 declaration

LEVEL 10

density

continuum_le_cardinalMk_dense_algebra_of_singleton_of_not_isCompactOperatorModel

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.continuum_le_cardinalMk_dense_algebra_of_singleton_of_not_isCompactOperatorModel

Exact source-only density bridge declaration; no Declaration Card is created.

IMMEDIATE PREREQUISITES IN THIS PROJECT

[isCompactOperatorModel_of_singleton_of_dense_algebra](#)

USED BY IN THIS PROJECT

None in this Project

[Read exact source](#)

▼ Technical Project identity

```
{
  "approved_exposition_records_sha256": "11b6ea11f79cb3f71258d81e2f5a53ad2f9967251e30c3e2704ace5f1dd46af2",
  "approved_review_items_sha256": "377f2d01ab7b6aee493a488ddad5cd6a80036e18ce10e72272096ab63f6782fa",
  "boundary_inputs_all": 2690,
  "boundary_inputs_curated": 15,
  "card_ref_set_sha256": "cd0a1f07158ec83c8d9350b7e9b63f41e3fe29ac6eb138e8630c30e6439ee4c3",
  "display_edges": 88,
  "full_native_graph_sha256": "e0c69751546f310fb7e15d2915c7e7b0f6a710eb0bc33269b24c2b228bcb5425",
  "graph_portable_sha256": "ef9c1507b18a89758ab43fe2be5efb02144590a80ea64825f97431c19ee9c1ae",
  "maximum_level": 10,
  "public_cards": 0,
  "reachability_pairs": 509,
  "selected_cards": 33,
  "source_tag": "v0.4.0"
}
```

Project source and formal graph are bound to MathlibAnnex v0.4.0. Publication history and the outer site frame belong to the hosting release.